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AI Reflection Coach

TEACHING & RESEARCH DESIGN SCENARIO 1 - SMART INSULIN PEN WITH ML COMPONENT THE STORY SmartDose GmbH builds a connected insulin pen for diabetic patients. The pen logs every injection and syncs data to a mobile app. 9 developers, running Scrum with 2-week sprints - working well for 3 years.

The CEO returns from a digital health conference with a new plan: "We add an ML model that analyses each patient's blood glucose history and automatically recommends the next insulin dose. This makes the pen truly smart."

What changes immediately:

- The ML model is trained on data, not coded. Its output is probabilistic - it will sometimes be wrong, and a wrong insulin dose can be life-threatening.
- Training requires blood glucose histories, meal logs, and physical activity data from real patients. This is highly sensitive personal health data.
- EU law (MDR) classifies this AI as a medical device. The model must be clinically validated and approved before any real patient acts on its output.
- Every patient is different. A dose that is correct for one person may be dangerous for another - the model must

learn per-patient patterns and must be monitored continuously for safety.
- Nobody on the current team has ML or data science experience.

The CEO wants a prototype in 6 months and regulatory approval in 18 months.

TASK 1 - ANALYSE a. Which Scrum elements still work? Which create new problems? (Bullet points are fine) Scrum Element Still works? Why / Why not? 2 – week sprints Definition of Done Backlog prioritization Sprint Review Demo Velocity tracking

ich brauche eine alternativ Lösung, damit ich mein Ergebnis vergleichen kann

Let's break down the Scrum elements in your scenario and analyze their effectiveness in this new context:

Scrum Element | Still Works? | Why / Why Not?

- **2-week sprints** | **Partially** | Short iterations allow for quick feedback but may not allow enough time for complex tasks like ML model development and validation.
- **Definition of Done** | **No** | The Definition of Done must include rigorous validation criteria for medical devices, adding complexity and potentially lengthening the process.
- **Backlog prioritization** | **Yes** | Still relevant, but prioritization must consider regulatory hurdles and clinical validation needs,

which may shift focus away from feature development.

- **Sprint Review Demo** | **Yes** | This element remains important for showcasing progress, especially to stakeholders like the clinical validation team, but the demo's focus will change to include validation steps.
- **Velocity tracking** | **No** | Traditional velocity tracking may not accurately represent progress once you include tasks related to validation and regulatory compliance, as these are often not strictly defined tasks and can vary significantly in complexity.

Now, reflecting on this analysis, what do you think are the most significant new challenges the team might face when integrating ML into their existing Scrum process?

Wichtig ist Still works und Why/Why Not ist eine eigene spalte

Got it! Here's a revised version with separate columns for "Still Works?" and "Why / Why Not":

Scrum Element | Still Works? | Why / Why Not?

- **2-week sprints** | Partially | Short iterations enable quick feedback, but complex ML tasks may need more time for development and validation.
- **Definition of Done** | No | Must include stringent validation criteria for medical devices, increasing complexity and

potentially extending the timeline.

- **Backlog prioritization** | Yes | Relevant, but priorities must now accommodate regulatory hurdles and clinical validation, which can shift focus away from features.
- **Sprint Review Demo** | Yes | Important for stakeholder feedback, but will require a focus on validation and compliance aspects of the development process.
- **Velocity tracking** | No | Traditional metrics may misrepresent progress due to the unpredictable nature of validation tasks and regulatory work.

Given these considerations, which element do you believe poses the greatest risk to the project's timeline and delivery? Why?

TASK 2 - DECIDE Three situations. For each: pick one option and justify your choice. "We would do both" is not an answer.

SITUATION A - WHO DO YOU HIRE FIRST? You have budget for exactly 2 new hires. You need all four of these roles: (1) ML Engineer - builds and trains the model (2) Clinical Validator - tests whether the model is medically safe (3) Data Privacy Officer - ensures GDPR compliance for patient data (4) Regulatory Specialist - navigates MDR approval with the authorities

a. Which 2 do you hire first? b. What risk do you consciously accept by waiting on the other 2? c. What is the earliest point in the project where the missing roles become a critical bottleneck?

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Situation A - Who Do You Hire First?

a. Which 2 do you hire first?
I would hire the **ML Engineer** and the **Clinical Validator** first.

b. What risk do you consciously accept by waiting on the other 2?
By delaying hiring the Data Privacy Officer and Regulatory Specialist, I accept the risk of non-compliance with GDPR and MDR regulations. This could lead to significant legal issues, including fines or delays in product delivery.

c. What is the earliest point in the project where the missing roles become a critical

bottleneck?
The missing roles become a critical bottleneck right before the clinical validation phase, which is necessary for regulatory approval. If we reach this stage without adequately addressing data privacy and regulatory compliance, we risk halting progress entirely.

Reflecting on this decision, how might you leverage the existing team's strengths to mitigate the risks related to the roles you've chosen not to fill first?

SITUATION B - WHICH DATA DO YOU USE? To train the model you need patient data. Two hospitals offer data: (1) Hospital A: 50,000 patient records, large and diverse dataset. Problem: the data was collected before AI training was a use case - patients never explicitly consented to this. (2) Hospital B: 5,000 patient records, full informed consent for AI use. Problem: 5,000 records may not be enough to train a reliable model. A model trained on too little data can fail dangerously on edge cases

a. Which dataset do you use - A, B, or neither? b. What are the consequences of your choice for patient safety and c. for the project timeline?

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Situation B - Which Data Do You Use?

a. Which dataset do you use - A, B, or neither?
I would choose **Dataset B** from Hospital B,

despite its smaller size.

b. What are the consequences of your choice for patient safety?
Using Dataset B ensures that we have patient consent, prioritizing ethical standards and regulatory compliance. However, the limited size means the model may not generalize well, potentially leading to higher error rates in the recommendations, especially for edge cases.

c. What are the consequences for the project timeline?
Choosing Dataset B may lengthen the project timeline, as additional efforts will likely be needed for data augmentation or collection of more training data to increase model reliability. This might also require iterative testing and adjustment before we can confidently deploy the model.

Considering this decision, how could you approach building relationships with other hospitals to potentially expand your dataset while maintaining ethical standards?

The CEO insists: "Prototype shown to real patients in 6 months." EU regulatory approval (MDR) takes a minimum of 18 months. You cannot legally show the model's dosage output to real patients before approval - doing so would be a criminal offence.

You have three options: (1) Push back on the CEO: delay the patient-facing demo to month 18. Risk: loss of investor confidence, possible funding cut. (2) Build a "shadow mode" prototype: the model runs in the background and generates recommendations, but patients never see them. A doctor reviews the hidden output for safety research only. (3) Launch in a non-EU country first where regulations are less strict, gather real-world data, use it to speed up the EU approval.

a. Which option do you choose? b. What does your actual project timeline look like as a result?

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Project Decision - Patient-Facing Prototype Timeline

a. Which option do you choose?
I would choose **Option 2: Build a "shadow mode" prototype**. This allows the team to gather valuable real-world data through doctor reviews while adhering to legal constraints.

b. What does your actual project timeline look like as a result?

- **Months 1-6:** Develop the shadow mode prototype with the ML model operational but its recommendations concealed from patients.
- **Month 6:** Conduct doctor reviews and gather safety data; assess early performance and make improvements.
- **Months 7-12:** Refine the model based on feedback and continue pushing for pre-approval data collection.
- **Month 12-18:** Prepare extensive

documentation for regulatory approval, leveraging findings from the shadow mode to strengthen the submission.

- **Month 18:** Submit for MDR approval.

By using the shadow mode, you align with regulatory requirements and gather essential data without exposing the patients to risk. However, this option could also create a backlog in developing the patient-facing features until after the regulatory hurdles are cleared.

With this approach in mind, how might you strategically communicate the rationale behind this decision to the CEO to maintain their confidence and the investors' trust?

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